

Open Items Register

Technical Design Document — Cubli

Armonia — Space Challenges 2026

Scan of August 2, 2026

How to use this document

This register lists every unresolved item in the TDD: the **TBD** markers, the sections that are empty, the subsections that exist only as comments, and the figures that are referenced but not drawn. It is a working checklist, not part of the submitted document.

Three status labels are used throughout:

- **READY** — decidable now. Nothing upstream is missing; the item is waiting on a decision or a datasheet lookup, not on data.
- **WAITING** — blocked on an upstream item, named in the row.
- **BLOCKER** — blocks a large number of downstream items. Close these first.
- **CLOSED** — resolved since the original scan. Retained in the register so the closure is auditable rather than silently deleted.

Section ?? states the dependency chain, which is what makes the ordering non-arbitrary: roughly a quarter of the register closes automatically once the packaging study closes.

1 Dependency chain

Most items are not independent. The governing chain is

$$V_{\min} \rightarrow L \rightarrow M \rightarrow h_w \rightarrow I_{w,\text{target}} \rightarrow \text{wheel design, feasibility gate, test thresholds}$$

so the packaging study (item **A4**) is the true head of the register. The parametric calculator `analysis/SIZING.py` implements Stages 1–6 of the closure procedure and now reports a converged fixed point at $M = 1.7716\text{ kg}$, reproduced by its own Stage 6 roll-up to within 0.1 g. The figures below have been **transcribed into the TDD** (A1, A2, A8 and parts of A4/A5 are closed). They remain provisional because roughly 34% of the mass total is still structural allowance rather than measured mass.

The three script-versus-document conflicts are now closed. Resolved as follows, and the document has been populated accordingly.

1. **$\omega_{\max}$: resolved to 6000 rpm**, matching the TDD and `electrical/MOTOR_SPEED.md`; the script's code has been corrected from 5000 rpm. 5000 rpm is retained as a documented over-dimensioning check — it raises $I_{w,\text{target}}$ by 6/5, so a wheel closing there closes at 6000 rpm with margin. A footnote to `tab:jumpup_budget` records this.
2. **Wheel geometry: resolved.** `tab:rw_params` now carries the as-designed ring — 120 mm OD, 15 mm radial width, round M6 through-holes, $N = 15$ on $R_{\text{pcr}} = 52.5\text{ mm}$, $m_w = 172.5\text{ g}$,

Table 1: Converged values from `analysis/SIZING.py`, as transcribed into the TDD. The wheel diameter is pinned by the packaging envelope (`D_w_FIXED_mm = 120`), not chosen by the mass-minimising sweep.

Symbol	Quantity	Converged value	Status
M	Cube mass (Stage 1 assumption)	1.7716 kg	Fixed point
M_{est}	Cube mass (Stage 6 roll-up)	1.7716 kg	$\Delta \approx 0$; 34% allowance
L	Cube edge length	150 mm	Fixed
η	Energy margin	1.05	Chosen
τ_b	Brake torque per wheel	5 N m	Assumed
$\tau_g = MgL/2$	Gravitational tip-over floor	1.303 N m	Derived
β	Brake-amplification factor	1.353	Derived
ω_{max}	Max wheel speed	6000 rpm	Agreed with TDD
h_w	Required angular momentum	$0.2588 \text{ kg m}^2 \text{ s}^{-1}$	Derived
$I_{w,\text{target}}$	Target wheel inertia (corner)	$4.119 \times 10^{-4} \text{ kg m}^2$	Derived
$I_{w,\text{target}}$	Target wheel inertia (edge)	$3.737 \times 10^{-4} \text{ kg m}^2$	Not critical
D_w	Wheel outer diameter	120 mm	Envelope constraint
N	Ballast stations per wheel	15 of $N_{\text{max}} = 39$	Derived
m_w	Wheel mass (each)	172.5 g	Derived
I_w	Wheel inertia achieved	$4.599 \times 10^{-4} \text{ kg m}^2$	111.7 % of target

$I_w = 4.599 \times 10^{-4} \text{ kg m}^2$. The diameter is an envelope constraint: left free, the mass-minimising sweep prefers a larger ring that does not fit the cube, at a cost of 85 g per wheel.

- 3. Ballast retention: resolved.** The blind-pocket language has been removed from `06_testing.tex` (part checks) and `09_cad.tex` (print settings); the through-bolt in tension is the stated retention feature, at 1.47 % of M6 class 8.8 proof load and 4.02 MPa bearing stress.

What remains open on the sizing chain. The residual risk is not a conflict but a margin asymmetry: the wheel sits 11.7 % above target while one station is worth 5.9 %, so downward ballast trim is nearly exhausted (−1 station) while upward trim is ample (+24). A structure that comes in *lighter* than the 610 g allowance is therefore the awkward case, and ω_{max} is the relief variable.

2 TBD items, by owning table

93 of the 97 TBD markers sit in tables. Each block below is one table. Items marked **CLOSED** have been populated since the original scan and are retained for audit; 78 markers remain open.

2.1 A1 — Jump-up sizing budget **CLOSED**

tab:jumpup_budget, sections/04_modeling_control.tex. **6 TBD — all populated** from the converged run of Table ???. The values remain provisional in the sense that M carries a 34% structural allowance, but they are no longer missing and the table is self-consistent with Stage 6.

Line	Symbol	Quantity	Closed by
189	M	Cube mass	CAD mass roll-up
191	η	Energy margin	Design choice (1.02–1.05)
193	$\tau_g = MgL/2$	Gravitational tip-over floor	Derived from M , L
194	β	Brake-amplification factor	Derived from τ_b , τ_g
196	h_w	Required angular momentum	Derived, Eq. (hw_ideal)
197	$I_{w,target}$	Target wheel inertia	Derived, Eq. (Iw_target)

Four of the six are pure functions of M and L , so this whole block closes the moment the mass roll-up is real rather than allowance-based.

2.2 A2 — Required inertia by contact case **CLOSED**

tab:inertia_cases, sections/05_hardware_design.tex. **4 TBD — all populated**. The calculator now evaluates both contact cases rather than only the sizing one: edge $h_w = 0.2348 \text{ kg m}^2 \text{ s}^{-1}$, $I_{w,target} = 3.737 \times 10^{-4} \text{ kg m}^2$; corner 0.2588 and 4.119×10^{-4} respectively. The corner case demands 9.3% more inertia, which is now shown rather than asserted.

2.3 A3 — Rail current budget **READY**

tab:power_budget, sections/05_hardware_design.tex, lines 461–469. **14 TBD**: typical and worst case for each load.

Line	Load	Governing figure
461	Teensy 4.1	Datasheet
462	XIAO ESP32-C6 telemetry	TX burst ($\approx 300 \text{ mA}$) governs
463	BMI270 IMU	Datasheet
464	MG92B brake servo	Stall current governs
465	MA600 encoders (3×)	Datasheet
467	Logic rail total	Checked against LM2596 rating (derates without heatsink)
469	moteus-n1 drivers (3×)	Spin-up, not balance, governs

Typical values are datasheet lookups and can be filled today. Worst case is explicitly defined in the TDD as the *coincident* case — telemetry burst, servo stall and full sensor load simultaneously — not the sum of typicals.

2.4 A4 — Driving CAD parameters **BLOCKER**

tab:cad_params, sections/09_cad.tex. **6 TBD, unchanged.** L , t_w , D_w and t_r are populated: D_w has been corrected from 13 cm to the 120 mm of the as-designed wheel, and the axial thickness $t_r = 5$ mm added, since the CAD needs it and the sizing fixes it. The six remaining TBDs are genuinely open — they need the packaging study, not the calculator, and the sizing work does not touch them.

Line	Symbol	Parameter
34	d_m	Motor axis offset from cube centre
35	h_m	Motor mounting face height on its axis
36	c_w	Wheel-to-subframe axial clearance
37	c_o	Clearance between orthogonal wheel modules
38	$V_{\min}$	Minimum internal clear volume
39	r_c	Corner contact feature radius

$V_{\min}$ is step 2 of the closure procedure and therefore the head of the whole dependency chain: it fixes L , which fixes M , which fixes $I_{w,\text{target}}$. c_w and c_o are additionally consumed by the interference checks of the CAD verification stage.

2.5 A5 — CAD model tree and part list **READY**

tab:cad_parts, sections/09_cad.tex, lines 65–89. **35 TBD:** fabrication route and status for every part, plus five missing quantities.

Line	Ref	Part	Missing
65	M1	Motor subframe	Qty, route, status
66	M2	Motor mount ($\times$ axis)	Route, status
67	M3	Battery cradle / retention	Route, status
68	M4	Electronics tray	Route, status
69	M5	Brake servo bracket	Route, status
73	W1	Ballasted ring	Route, status
74	W2	Wheel hub / rotor adapter	Route, status
75	W3	Ballast bolt and nut set	Qty, status
76	W4	Brake strike feature	Route, status
80	F1	Frame member / edge rail	Qty, route, status
81	F2	Corner contact feature	Route, status
82	F3	Face panel / wheel guard	Qty, route, status
86	R1	Face plate (1-DoF rig)	Route, status
87	R2	Pivot bearing mount	Route, status
88	R3	Rig motor mount	Route, status
89	R4	Hard stops / containment	Qty, route, status

W3's quantity is $3N$ where N is the ballast count. With the wheel geometry now settled at $N = 15$ per wheel, this has been committed as **45** M6 bolt-and-nut sets, leaving 34 TBD in this table. The rest are fabrication routes and statuses, which are decidable today and do not depend on the sizing.

2.6 A6 — Print and manufacturing settings **READY**

tab:print_settings, sections/09_cad.tex, lines 138–143. **18 TBD:** material, infill/walls and orientation for W1, W2, M1, M2, F1, F2 — the six structural printed parts.

The TDD states the requirement these settings must meet: the wheel ring carries centrifugal load at the ballast holes and the motor mounts carry the brake impulse, so orientation is specified relative to the principal load direction and a part reprinted differently is not the part that was analysed. The calculator assumes PET-CF at 1290 kg m^{-3} , which should be the material entry for W1 at minimum.

2.7 A7 — Mass properties verification **WAITING**

`tab:mass_verification`, `sections/09_cad.tex`, lines 197–200. **12 TBD**: CAD value, measured value and discrepancy for total cube mass M , wheel mass m_w , wheel inertia I_w and CoM offset.

The CAD column can be filled as soon as the model exists; the measured column requires the assembled hardware (assembly step 8). This is the table that closes the mass loop, so a discrepancy here re-enters $I_{w,\text{target}}$.

2.8 A8 — Pre-fabrication feasibility gate **CLOSED**

`tab:jumpup_gate`, `sections/06_testing.tex`. **8 TBD** — **all populated**, and the gate passes on every row: $\omega_w = 5374 \text{ rpm}$ against $\omega_{\text{max}} = 6000 \text{ rpm}$ for a speed margin of 0.896; $I_w = 4.599 \times 10^{-4}$ against $I_{w,\text{target}} = 4.119 \times 10^{-4}$, a 11.7% route disagreement; trim authority $-1/+24$ stations; retention at 1.47% of proof load. Row 116 has also been reworded off the superseded blind-pocket scheme, as flagged below.

Line	Quantity	Pass condition
109	Required wheel speed ω_w	$\leq$ achievable speed with margin
110	Achievable wheel speed ω_{max}	Sustained in service, not no-load
111	Speed margin $\omega_w/\omega_{\text{max}}$	< 1 by a stated margin
112	Target inertia $I_{w,\text{target}}$	—
113	Achieved inertia I_w	$\geq I_{w,\text{target}}$
114	Route disagreement	Reported; explained if large
115	Ballast trim authority	Non-zero in both directions
116	Nut retention margin	Retention load carried by a stated feature

Row 116 was written against the superseded blind-pocket scheme; it has been reworded to “Ballast retention margin” with the pass condition stated on bolt tension and head/nut bearing stress, which is what the through-hole design actually relies on. The same correction was applied to the part-check paragraph of `06_testing.tex` and the print-settings preamble of `09_cad.tex`, both of which still referred to pockets and a retention floor.

2.9 A9 — 1-DoF plant identification **WAITING**

`tab:paramid_methods`, `sections/06_testing.tex`, lines 307–310. **4 TBD**: torque constant K_t , body inertia J , friction, loop latency. Methods and instruments are already specified — only the measured values and their tolerances are pending. Requires the rig (gate M6).

2.10 A10 — Control performance metrics **WAITING**

`tab:perf_metrics`, `sections/06_testing.tex`, lines 405–410. **5 TBD**: targets for recovery angle, settling time, disturbance-rejection bandwidth, slew tracking error, momentum drift. Control effort is already defined by reference to the driver limits and carries no TBD.

2.11 A11 — Connector schedule **READY**

`tab:connectors`, `sections/10_electrical_detail.tex`, lines 78–88. **11 TBD**: housing and keying for J1–J11.

The governing rule is already stated in the TDD: no two connectors carrying different voltages may share a housing type, so a mis-mate is impossible by construction. J1 is fixed by the BOM (XT90). The remaining ten are open choices and can be decided today.

2.12 A12 — Wire schedule READY

tab:wire_schedule, sections/10_electrical_detail.tex, lines 112–118. **14 TBD**: gauge and length for H1–H7. The sizing basis is already stated per row (peak bus current for H1–H2, per-axis spin-up for H3, motor phase current for H4, topology for H5, length limit for H6, stall current for H7). Gauge follows from A3; length follows from the CAD envelope.

2.13 A13 — As-built electrical measurements WAITING

tab:electrical_measured, sections/10_electrical_detail.tex, lines 200–206. **7 TBD**: K_m , I_w from spin-down, viscous and Coulomb friction, command-to-encoder latency, achievable wheel speed on the bus, worst-case logic rail load, runtime per charge. Each row already names the estimate it replaces. Requires hardware.

3 TBD in prose

Four TBD markers sit in running text rather than in a table.

Location	Pending
06_testing.tex:12	Generic statement that thresholds are TBD. Resolves itself when the tables close; no separate action.
06_testing.tex:202	Robustness sweep magnitude — how far plant parameters are perturbed about nominal. Set once the identification tolerances of A9 are known.
06_testing.tex:355	Edge-balance numeric targets beyond the G6 gate criterion.
06_testing.tex:369	Slew tracking targets.

4 Empty sections

Section	File	State
Appendix D — Control Algorithms	11_control_algorithms.tex	BLOCKER 0 bytes. Input by main.tex:151 and cross-referenced six times as app:control_algorithms from Sec. 5 and Sec. 6
§7.1 NMPC & warm-start control	07_future_work.tex:4	Heading only
§7.2 Autonomous stand-up	07_future_work.tex:6	Heading only
§7.4 Single-wheel variant	07_future_work.tex:262	Heading + comment

Appendix D is the most serious content gap in the document: six live cross-references point into an empty file. The TDD promises it will contain the mode state machine, the estimator and controller in executable pseudocode, the fixed gains and tuning parameters, the timing budget and the fault responses.

The three empty Future Work headings should either be written or deleted — empty subsections in a submitted document read worse than a shorter list.

5 Comment-only stubs

Twenty subsections exist only as LaTeX comments describing what should be there.

Section 3 — System Analysis & Requirements **BLOCKER**

Three of four subsections are stubs. At roughly 230 words this section is a hole in the middle of the document and is more conspicuous to an assessor than any amount of over-length prose elsewhere.

Line	Subsection	What the comment asks for
7	§3.1 Scope of work & use cases	Boundaries of the design effort; the four target behaviours as primary use cases
14	§3.2 Actuation trade-off	RW-vs-CMG justification. Prose exists; the comment is unresolved
48	§3.4 Coordinate frames & conventions	Body frame, world frame, sign conventions for tilt and wheel angle

Mechanical

Location	Content
05_hardware_design.tex:204	2D prototype structural concept
05_hardware_design.tex:586	§5.5 System Integration — “TBD pending CAD”
09_cad.tex:97, 101	2D prototype construction notes; rig general-assembly figure
09_cad.tex:107, 111, 112	3D cube drawings; exploded view; section view through two orthogonal wheel modules

Electrical — eleven figure placeholders

`sections/10_electrical_detail.tex`: schematic sheets 1–7 at lines 18, 25, 32, 38, 44, 50 and 55; harness drawing and as-built photograph at lines 123 and 125; perfboard floorplan and board outline drawing at lines 130 and 133.

Each placeholder already names the constraint the sheet must be verified against, so the drawing brief is written — only the drawings are missing.

6 Missing figures

Twenty-one figures are referenced or described but not present: the twenty placeholders above, plus one more.

- `04_modeling_control.tex`:21–22 — free-body diagram of the 2D pendulum, labelling θ_b , θ_w , l , l_b and the pivot. This is the only figure suggested in Sec. 4, and that section currently has no figures at all.

7 Defects found alongside the TBDs

These are not open items in the same sense — they are errors in content that is already written, and each is a short fix.

#	Location	Defect
1	05_hardware_design :420 vs 08_bom:86	Battery capacity contradiction: 3600 mA h in the power architecture, 1550 mA h in the BOM, same 6S pack. The sizing script uses 3600.
2	05_hardware_design	CLOSED <code>tab:rw_params</code> was missing its <code>\caption</code> and would not have appeared in the List of Tables. A caption is now present.
3	06_testing:65–66 vs <code>tab:gates</code>	G6 hard-kill date stated as 13 August; <code>Table tab:gates</code> puts G6 at 17 August. A kill date cannot precede the gate it kills.
4	02_project_organization _tasks:46	Task B4 says “Raspberry Pi software architecture”; Sec. 5 and the BOM specify Teensy 4.1 throughout.
5	08_bom:64 vs <code>tab:cad_parts</code>	Brake servo quantity: BOM says 1–3, the CAD part list says 3 brackets, Sec. 5 says “servos” plural.
6	Author list vs <code>tab:team_roles</code> vs task table	Name spelling: “Niccolò” / “Niccolo” / “Nicc”.
7	02_project_organization _tasks:79–82	Task IDs C11–C14 appear under the “3D Design” block, after D1–D7.
8	All section files	British/American spelling mixed: realise/realize, characterise/characterize, behaviour/behavior, sometimes within one section.
9	04_modeling_control:131	Hard-coded <code>\newpage</code> splits a sentence across pages mid-argument.
10	All tables	CLOSED Every float used <code>[h]</code> ; all are now <code>[htbp]</code> . No <code>[h]</code> float remains in <code>sections/</code> .
11	sections/	<code>code.cpp</code> and <code>code.exe</code> are untracked in the repo; a compiled binary should not be committed.

8 Suggested order of work

1. **CLOSED** The three script-versus-document conflicts of Section ?? are resolved — $\omega_{\max}$ at 6000 rpm, the wheel table rebuilt on the as-designed 120 mm through-hole ring, and the blind-pocket retention argument removed. A1, A2 and A8 were then populated from the converged run (19 TBD closed).
2. **Close the packaging study** (A4) — now the head of the register. $V_{\min}$ and the clearances release L , then M . Note that A1, A2 and A8 are populated but *provisional*: they rest on a 610 g structural allowance, so closing A4 will move them again. That is the next turn of the fixed-point iteration, not a re-opening of settled items.
3. **Write Appendix D**, or remove the six cross-references that point into it.
4. **Write Section 3**, which is three-quarters empty.
5. **Do the READY items**, which need no upstream data: A3 (rail budget), A5 (fabrication routes), A6 (print settings), A11 (connectors), A12 (wire gauges).
6. **Fix the eleven defects**, which cost credibility out of proportion to the effort of fixing them.
7. **The remaining WAITING items** (A7, A9, A10, A13 and the four prose TBDs) close on hardware and cannot be accelerated on paper.